AI Clinical Decision Support Lite Hackathon

Day 1

Research, Scope & Document Ingestion

AI Clinical Decision Support Lite Hackathon

Understand
Clinical RAG

→

Decide
Scope + sources

→

Build
First index

What students will learn and execute during Day 1

Today’s learning objectives

Students should leave this session ready to implement the first working knowledge base.

INSTANT

Conceptual goal

Understand why clinical RAG is different from a normal chatbot: the answer must be grounded in official medical evidence, not model memory.

Technical goal

Learn the first pipeline: PDF parsing, cleaning, section-aware chunking, metadata preservation, embeddings, and vector indexing.

Execution goal

Finish the day with selected guidelines, a narrow scope, and a searchable knowledge base tested with sample queries.

Why this hackathon matters

The challenge is not fluent text. It is safe, traceable, evidence-based clinical guidance.

A normal LLM may answer confidently even when it has no evidence.

Student takeaway

If the system cannot find evidence in the uploaded guidelines, it should say so instead of guessing.

Fluent answer ≠ Safe answer → **Retrieved evidence + Citation** → **Grounded response**

Day 1 foundation

Every later score depends on today's foundation: useful documents, clean chunks, and correct citation metadata.

What are we building?

A lightweight clinical decision-support assistant that answers from official guidelines only.

PDF Guidelines → **Parse + Chunk** → **Embed + Index** → **Retrieve Top-K** → **Generate Answer** → **Citations + Safety**

Input

Official medical guideline PDFs and a clinical question.

Core logic

Search the guidelines, select relevant chunks, then answer only from them.

Output

Recommendation, supporting evidence, citation, confidence, and disclaimer.

Day 1 roadmap

The day moves from problem definition to the first searchable guideline index.

1 → 2 → 3 → 4 → 5

1. Scope

Pick a narrow clinical problem.

2. Sources

Select official, legally usable PDFs.

3. Parsing

Extract clean text from each PDF.

4. Chunking

Split by sections and preserve context.

5. Indexing

Generate embeddings and store metadata.

End-of-day target

A demo-ready knowledge base is not required today. A working first index with visible chunks and metadata is required.

Step 1 — Choose a narrow clinical scope

The best scope is small enough to evaluate and important enough to demonstrate.

Good examples

• Adult hypertension initial management.
• Diabetes screening guidance.
• Asthma step-up / step-down guidance.
• Preventive screening recommendations.
• Antibiotic use for one condition area.

Avoid today

• “All diseases assistant.”
• Mixing too many specialties.
• Rare cases with weak guideline coverage.
• Patient-specific diagnosis.
• Emergency medical advice.

Scope statement template

This system helps [target user] answer questions about [clinical topic] using [official guideline source].

Team task: write your scope statement in one sentence before touching code.

Step 2 — Select credible guideline PDFs

Your dataset must be trusted, legal, public, and manageable.

Allowed data

• Official medical guideline PDFs.
• Public health documents.
• Publicly accessible recommendations.
• Moderate-size collections.
• Sources that can be cited clearly.

Not allowed

• Hospital records.
• Private patient data.
• PHI or identifiable health data.
• Credential-gated datasets.
• Extremely large research corpora.

Suggested sources

• WHO clinical guidelines.
• CDC recommendations.
• NICE guidelines.
• USPSTF recommendations.
• Other official bodies with clear public use.

Step 3 — Build the ingestion pipeline

Turn guideline PDFs into clean, searchable, citable data.

PDF Parsing → Text Cleaning → Section Detection → Chunk Creation → Metadata Storage

PDF parsing

Extract text while preserving page numbers. Check if tables and multi-column layouts are readable.

Cleaning

Remove repeated headers, footers, page noise, broken line breaks, and obvious extraction artifacts.

Section detection

Keep headings and recommendation sections because they make retrieval and citations stronger.

Step 4 — Section-aware chunking

A good chunk contains one coherent clinical idea and enough context to support an answer.

Chunking decision

Too small
Missing context

→

Balanced
One useful idea

→

Too large
Low precision

Recommended start

Use section-aware chunks around 400–800 tokens with overlap only when needed.

Preserve context

Keep section title, recommendation label, page number, and document name with every chunk.

Manual check

Open 5 chunks and verify that each one makes sense without the entire PDF.

Step 5 — Metadata is the citation foundation

Citations shown later are only as accurate as the metadata saved today.

Minimum chunk object

```
{
  "document_name": "NICE Hypertension
Guideline",
  "page_number": 18,
  "section_title": "Starting treatment",
  "chunk_id": "nice_htn_p18_c03",
  "source_url": "...",
  "text": "..."
}
```

Why it matters

Without document name, page, and section, the final answer may look cited but cannot be trusted.

Student task

Store metadata with the vector entry, not in a separate spreadsheet that can get out of sync.

Quality check

Pick one chunk and trace it back to the original PDF page.

Step 6 — Build the first vector index

The goal today is not perfect retrieval. It is a working baseline that can be improved tomorrow.

Chunk Text → Embedding Model → Vector Database → Top-K Search Test

Embedding choice

Use a reliable text embedding model. Keep the model name and settings documented.

Index storage

Use any vector database, but make sure scores and metadata can be retrieved with the text.

Baseline test

Run 5–10 clinical questions and inspect the returned chunks before generation.

Hands-on lab — Complete this today

Practical checklist for teams after the teaching block.

- Choose one narrow clinical topic
- Download 1–2 official guideline PDFs
- Document why the source is credible and public
- Extract text and inspect sample pages
- Create section-aware chunks
- Attach metadata: document, section, page, chunk ID
- Generate embeddings and index chunks
- Run 5–10 clinical queries and display chunks

Stop point: Do not optimize prompts or UI until the first index can retrieve reasonable evidence.

End-of-day review questions

Use these questions to decide whether the team is ready for Day 2: Retrieval Optimization.

Source readiness

- Are the PDFs official and public?
- Is the scope narrow?
- Are source URLs and names recorded?
- Are licensing concerns avoided?

Chunk readiness

- Are chunks readable?
- Do chunks preserve page and section?
- Are recommendations split correctly?
- Is garbage text removed?

Search readiness

- Are embeddings created?
- Is metadata returned with chunks?
- Do 5–10 questions retrieve reasonable evidence?
- Are weak results visible?

Ready for Day 2 when the team can show: question → top chunks → source page/section.